AP PRECALCULUS

Unit 1 • Lesson 1.14 notes

Lesson 1.14 • Function Model Construction and Application

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Build a function model from a geometric or verbal description.
- Build a model from data with regression and use it to predict.
- Interpret model outputs in context, with units.

Vocabulary. regression, constraint, optimization, interpolation, extrapolation.

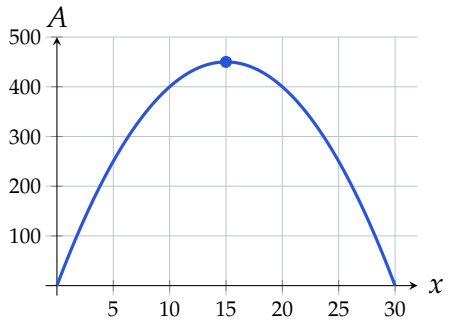
1. Building a model from a description

Name the variables, write the quantity of interest using one input variable (use the constraint to eliminate the others), and give the domain.

Example 1 • Fencing

A farmer has 60 m of fence for a rectangular pen along a wall (no fence needed on the wall side). Model the area.

Solution. Let x be the length of each side perpendicular to the wall. The side parallel to the wall is $60 - 2x$. Area: $A(x) = x(60 - 2x)$, domain $0 < x < 30$. $A(x) = -2(x - 15)^2 + 450$, so the maximum area is 450 m^2 when $x = 15$ m.



2. Building a model from data

With a graphing calculator, enter the data and choose the regression that matches the pattern (Lesson 1.13). Report the equation with rounded coefficients and use it within the range of the data when possible.

Example 2 • Regression

A ball's height (m) at times t (s): $(0, 1.2)$, $(1, 6.1)$, $(2, 9.0)$, $(3, 9.9)$, $(4, 8.6)$.

Solution. Quadratic regression gives $h(t) \approx -1.029t^2 + 5.974t + 1.183$. Prediction: $h(5) \approx 5.3$ m. The model's maximum is about 9.9 m at $t \approx 2.9$ s.

Common mistakes

- Leaving two variables in the model (use the constraint to write one in terms of the other).
- Giving an answer without units or context.

AP exam tip

On calculator questions, store regression coefficients in the calculator instead of retyping rounded values, and round only the final answer to three decimal places.

Quick check

1. A box has a square base of side x and volume 32. Write its height $h(x)$.
2. What is the domain of $A(x) = x(60 - 2x)$ in Example 1, and why?

Answers. 1. $h(x) = \frac{32}{x^2}$, $x > 0$. 2. $0 < x < 30$: both x and $60 - 2x$ must be positive.